



BESS Equipment Availability Deep Dive

How to measure and compute Equipment Availability

Module 1 - BESS Availability Explained

 by Fred Rivollier

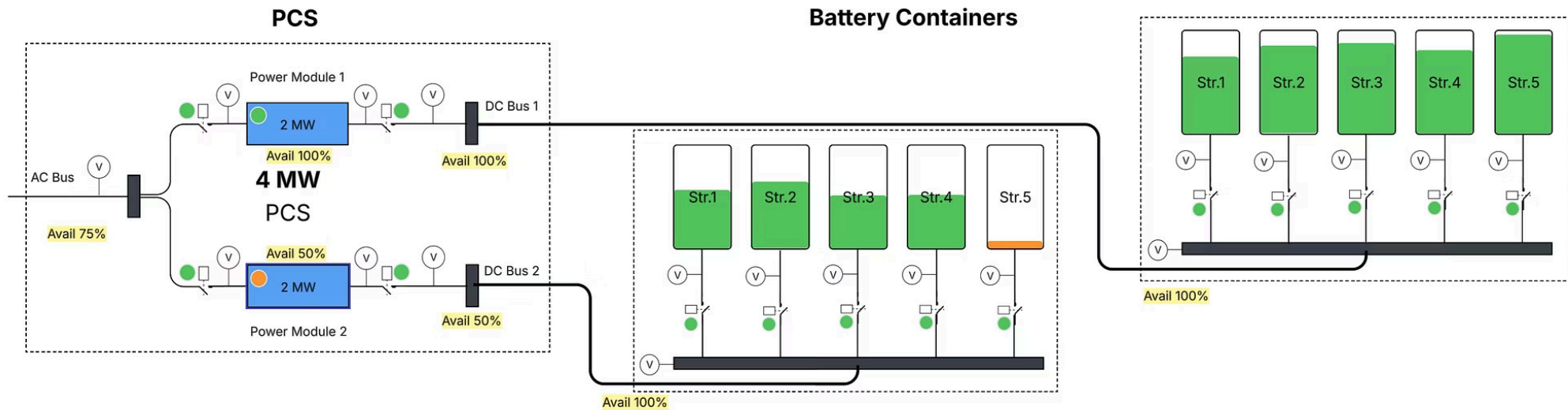
BESS Availability Types Recap

Combining all 4 availability type will form **OBE (Overall Battery Effectiveness)**

<u>Availability Type</u>	<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
① Equipment Availability	MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted
② Operational Availability	HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted
③ Power Availability	HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted
④ Energy Availability	HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	LOW to MEDIUM Calculated by EMS, include SoC Error and Imbalance	MEDIUM Might include power derated energy due to imbalance	MWh	%_installed %_contracted

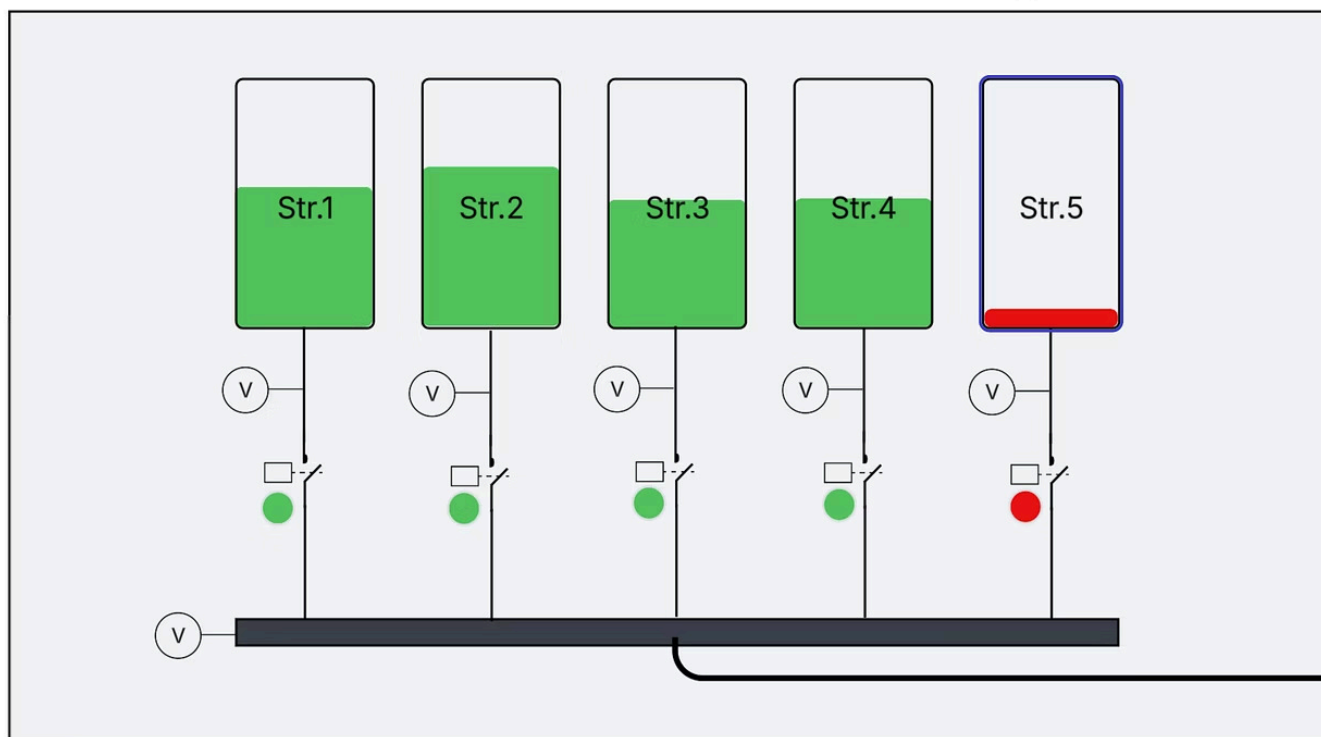
① Equipment Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted



How to measure Battery String Availability

Battery Container 5MWh (5 strings)



Measurements

Voltage
Sensor



e.g. 800V - 1400V

String BMS
Contactor



● Closed - Available

● Open - Unavailable

BMS Status



100% Available

0% Available

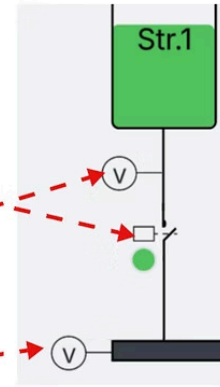
To PCS DC Bus

How to Compute Battery String Availability

1. From DC Contactor Status [**Open**, **Closed**]

2. Use Availability status from BMS [**0%**, **100%**]

3. Use Voltage Measurements for Quality Control:
 $\text{abs}(\text{String Voltage} - \text{Bus Voltage})$ [**>1V**, **<1V**]



**0% String
Availability**

5 MWh Battery Container



**100% String
Availability**

5 MWh Battery Container



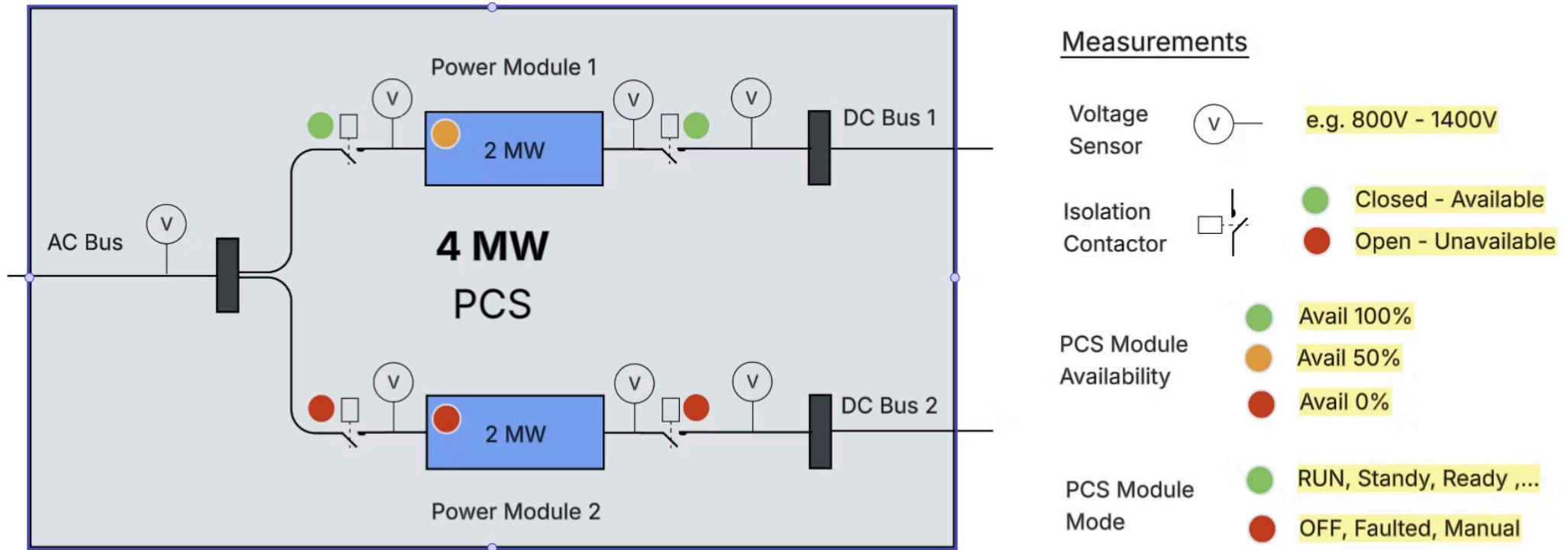
**80% String
Availability**

5 MWh Battery Container



String Avail = min(DC Contactor Status, BMS Status)

How to measure PCS Availability



How to Compute PCS Availability

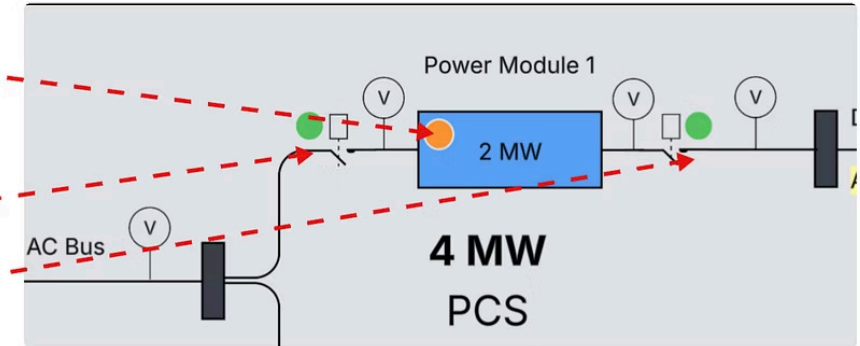
1. Use PCS provided Status [Ready, Run, STOPPED, FAULTED, OFF]

2. Use Availability Calculated by PCS [0%, 50%, 100%]

3. Verify Voltages accross Modules for quality control:

$\text{abs}(\text{AC PCS Voltage} - \text{Module AC Voltage})$ [$>1\text{V}$, $<1\text{V}$]

$\text{abs}(\text{DC Bus Voltage} - \text{Module DC Voltage})$ [$>1\text{V}$, $<1\text{V}$]



PCS Modules Avail = min(Module Status, Module Availability)

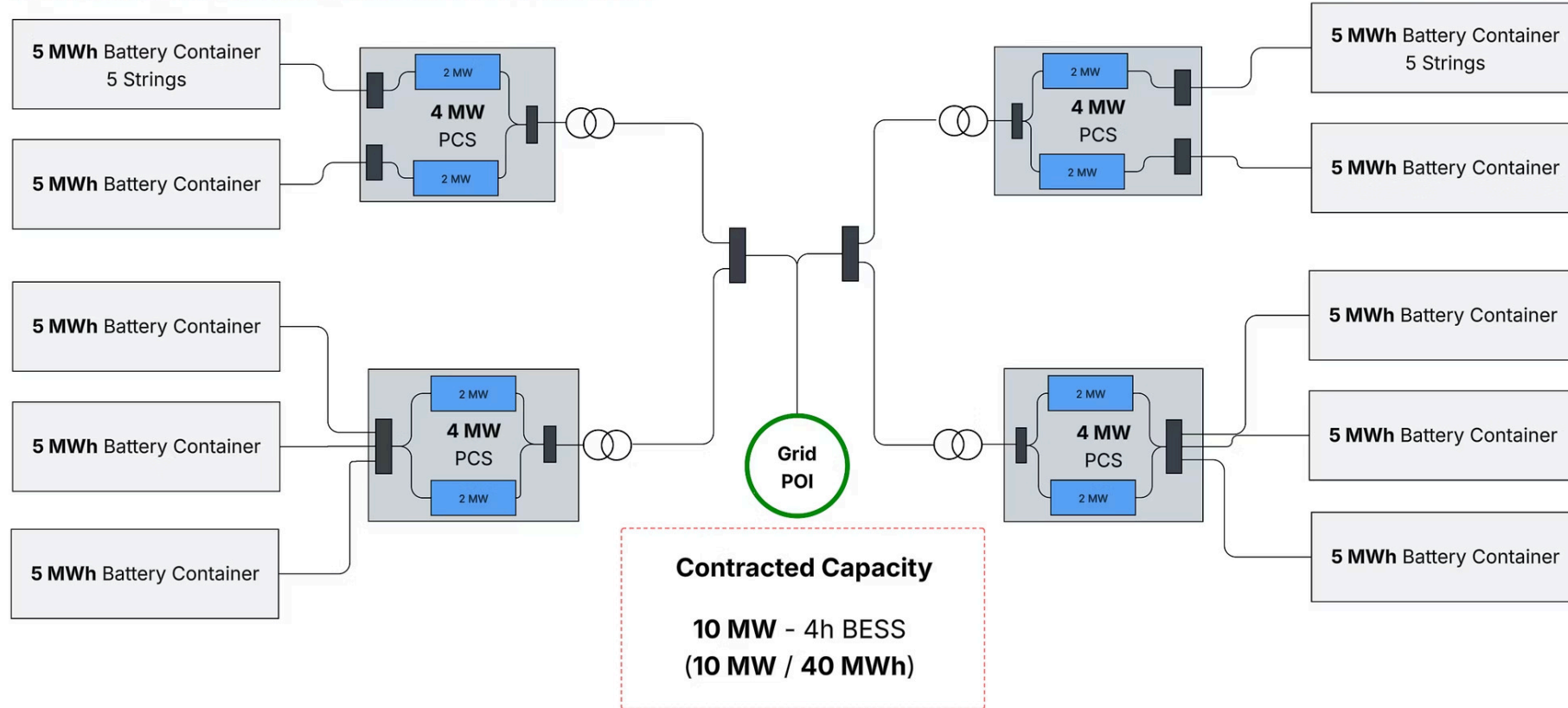
Example BESS

Installed Capacity

16 MW_{ac} / 50 MWh_{dc}
16 MW_{ac} / 45 MWh_{ac} (90% dc/ac efficiency)

4 PCS (8 PCS Modules)

6 Individual DC Bus
50 Strings



Normalization of Capacities for Availability Calculation

Contracted Capacity

10 MW - 4h BESS
(10 MW / 40 MWh)

Installed Capacity

16 MW_{ac} / 50 MWh_{dc}
16 MW_{ac} / 45 MWh_{ac} (90% dc/ac efficiency)

4 PCS (8 PCS Modules)

6 Individual DC Bus
50 Strings

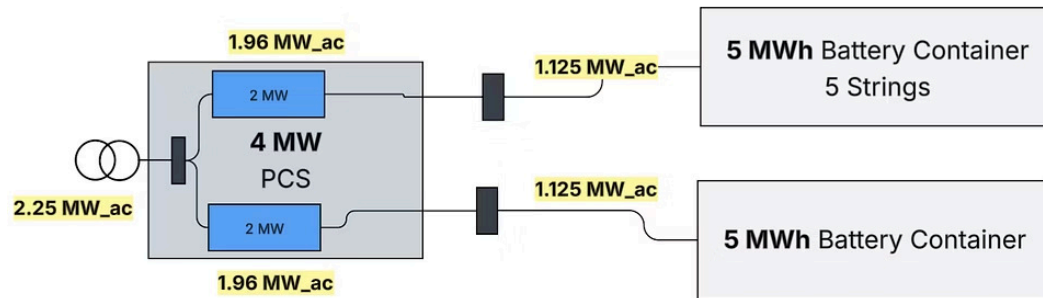
$$\text{Overbuild} = (45 - 40)/40 \text{ MWh} = 12.5\%$$

POI Normalized PCS Module Capacities to MW at POI:

PCS to POI losses have to be measured of estimated
est. to 2% loss for this example

$$1 \text{ PCS Module} = 4 \text{ MW} - \text{POI Loss} / 2$$

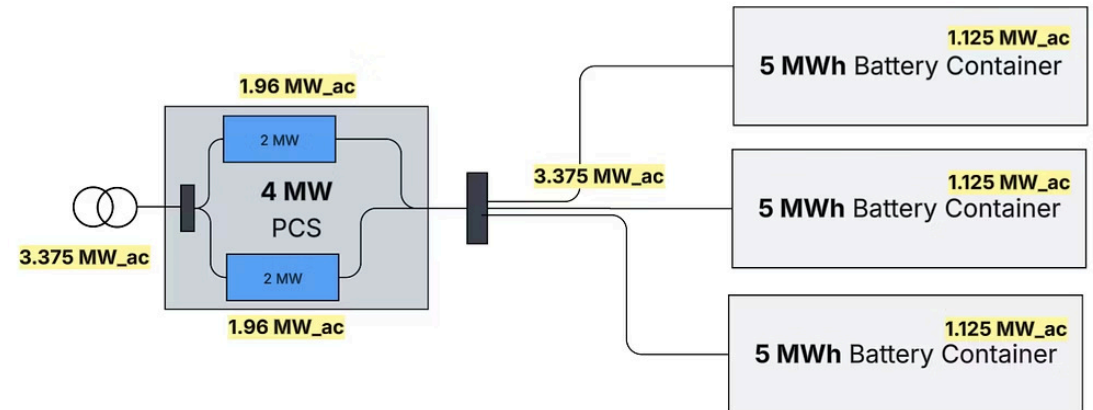
$$4 \text{ MW} * (1 - 0.02) / 2 = 1.96 \text{ MW/PCS Module}$$



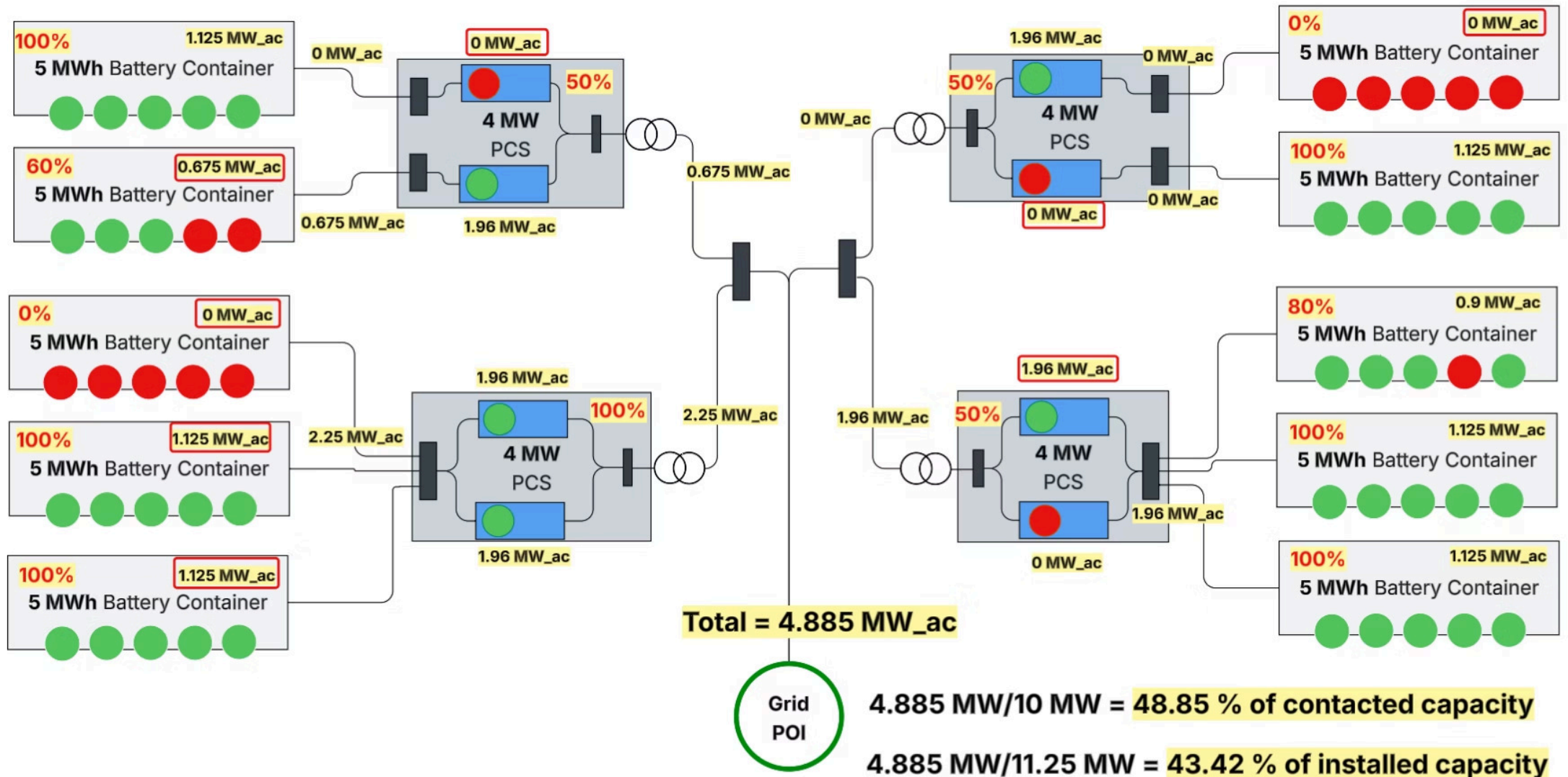
POI Normalized String Capacities to MW at POI:

$$1 \text{ String Energy Capacity} = 45 \text{ MWh} / 50 = 0.9 \text{ MWh/String}$$

$$1 \text{ String Power Capacity} = 45 \text{ MWh} / 4 / 50 = 0.225 \text{ MW/String}$$



Example BESS Equipment Availability



BESS Equipment Availability Formula

Dimensions and Definitions:

- ***bi*** = DC bus index
- ***si*** = string index within DC bus ***bi***
- ***i*** = calculation interval (1min - 5min)
- ***ep*** = evaluation period (1 day, 1 week, 1 month, 1 quarter, 1 year)
- ***N*** = number of calculation intervals ***i*** in the evaluation period ***ep***.
- **Excused_MW** = excused events quantified in MW per interval (schedule main, data loss while good availability, gred outages,)

Step 1 (Bus Level) (interval *t*): Availability_Bus_MW(*bi*,*i*) = $\min(\sum \text{Avail_String_MW}(\text{bi}, \text{si}, \text{i}), \sum \text{Avail_PCS_Mod_MW}(\text{bi}, \text{i}))$ if data available, else 0

Step 2 (BESS Level) (interval *t*):

Per Interval *i*

$$\text{Availability_BESS_MW}(i) = \sum \text{Avail_Bus_MW}(\text{bi}, \text{i}) + \text{Excused_MW}(i)$$

$$\text{Availability_BESS_Installed_}\%(i) = \min(\text{Availability_BESS_MW}(i) / \text{Installed_Capacity_MW}, 100\%)$$

$$\text{Availability_BESS_Contracted_}\%(i) = \min(\text{Availability_BESS_MW}(i) / \text{Contracted_Capacity_MW}, 100\%)$$

Step 3 (BESS Level) (eval period *ep*):

Per evaluation period *ep*

$$\text{Availability_BESS_MW}(ep) = \sum \text{Avail_BESS_MW}(i) / N$$

$$\text{Availability_BESS_Installed_}\%(ep) = \min(\text{Availability_BESS_MW}(ep) / \text{Installed_Capacity_MW}, 100\%)$$

$$\text{Availability_BESS_Contracted_}\%(ep) = \min(\text{Availability_BESS_MW}(ep) / \text{Contracted_Capacity_MW}, 100\%)$$



What's Coming Next

We've covered the Equipment Availability.

In our next sessions, we'll dive deeper into **Operational Availability**.

Stay tuned for more Performance Engineering insights!

